

ChatGPT

Dataset Summary and Format

CS-Aarhus_multiplex-Copy1.edgesFileDrosophila_multiplex-Copy1.edgesFilehomo_multiplex_2-Copy1.edgesFileKapferer-Tailor-Shop_multiplex-Copy1.edgesFilepierreauger_multiplex-Copy1.edgesFilesacchpomb_multiplex_2-Copy1.edgesFileVickers-Chan-7thGraders_multiplex-Copy1.edgesFilegive a summary of dataset what each one of them is about what is data in them in which format it is. 2

**CS-Aarhus_multiplex-Copy1.edges**

File

drosophila_multiplex-Copy1.edges

File

homo_multiplex_2-Copy1.edges

File

Kapferer-Tailor-Shop_multiplex-Copy1.edges

File

pierreauger_multiplex-Copy1.edges

File

sacchpomb_multiplex_2-Copy1.edges

File

Vickers-Chan-7thGraders_multiplex-Copy1.edges

File

give a summary of dataset what each one of them is about what is data in them in which format it is

Analysis paused

Here's a concise summary of each uploaded multiplex network dataset — what it represents, what kind of data it contains, and the file format.

General File Format

All datasets are stored in **edge-list format** (.edges files).

Each row has **4 columns**:

Column	Meaning
1	Layer ID
2	Source Node
3	Target Node
4	Edge Weight / Presence

Example:

```
1 10 14 1
```

means:

- Layer = 1
- Node 10 connected to Node 14
- Edge weight = 1

These are **multiplex networks**, meaning:

- Same nodes can interact in multiple ways
 - Each interaction type is represented as a separate layer
-

1. CS-Aarhus Dataset

File

CS-Aarhus_multiplex-Copy1.edges

About

A **social network** dataset from the Computer Science department at Aarhus University. It models different kinds of relationships between people.

Data Contains

Different layers represent different social interactions such as:

- Facebook friendship
- Leisure interaction
- Work collaboration
- Co-authorship
- Lunch interaction

Statistics

- **Layers:** 5
- **Nodes:** 61
- **Edges:** 620

Format Example

```
1 1 2 1
```

Node 1 connected to Node 2 in Layer 1.

Use Cases

- Social influence analysis
 - Community detection
 - Multiplex centrality
 - Information diffusion
-

2. Drosophila Dataset

File

`drosophila_multiplex-Copy1.edges`

About

A **biological multiplex network** based on the organism:

Drosophila melanogaster

This is commonly used in genetics and protein interaction studies.

Data Contains

Nodes represent:

- Genes or proteins

Layers represent different biological relationships such as:

- Genetic interactions
- Physical protein interactions

Statistics

- **Layers:** 2
- **Nodes:** 1363
- **Edges:** 3059

Use Cases

- Bioinformatics
- Gene interaction prediction
- Biological influence propagation

3. Homo Multiplex Dataset

File

`homo_multiplex_2-Copy1.edges`

About

A human biological interaction network.

Likely based on:

Homo sapiens

protein/gene interaction systems.

Data Contains

Nodes:

- Human proteins or genes

Layers:

- Different interaction types
 - Physical interactions
 - Functional associations
 - Co-expression relations

Statistics

- **Layers:** 3
- **Nodes:** 3471
- **Edges:** 23634

Special Note

This is the **largest dataset** among your uploads.

Use Cases

- Disease pathway analysis
- Protein interaction studies
- Biomedical network mining

4. Kapferer Tailor Shop Dataset

File

Kapferer-Tailor-Shop_multiplex-Copy1.edges

About

A classic sociology dataset collected by anthropologist:

Bruce Kapferer

It models social relations among workers in a tailor shop.

Data Contains

Nodes:

- Workers/employees

Layers:

- Different social relations:
 - Friendship
 - Work interaction
 - Conflict
 - Advice

Statistics

- **Layers:** 4
- **Nodes:** 39
- **Edges:** 1018

Use Cases

- Organizational behavior
 - Social conflict analysis
 - Small-group dynamics
-

5. Pierre Auger Dataset

File

`pierreauger_multiplex-Copy1.edges`

About

A scientific collaboration network related to the:
Pierre Auger Observatory

Data Contains

Nodes:

- Researchers/scientists

Layers:

- Different collaboration or communication types

Statistics

- **Layers:** 16

- **Nodes:** 514
- **Edges:** 7153

Special Feature

This dataset has the **highest number of layers**.

Use Cases

- Scientific collaboration analysis
 - Multilayer community detection
 - Research influence modeling
-

6. Sacchpomb Dataset

File

sacchpomb_multiplex_2-Copy1.edges

About

A biological network for:

Saccharomyces pombe

Data Contains

Nodes:

- Genes/proteins

Layers:

- Different biological interaction types

Statistics

- **Layers:** 3
- **Nodes:** 811
- **Edges:** 8363

Use Cases

- Cellular process analysis
 - Biological pathway modeling
 - Network biology research
-

7. Vickers-Chan 7th Graders Dataset

File

Vickers-Chan-7thGraders_multiplex-Copy1.edges

About

A school social network dataset involving 7th-grade students.

Data Contains

Nodes:

- Students

Layers:

- Different relationships such as:
 - Friendship
 - Working together
 - Social preference

Statistics

- **Layers:** 3
- **Nodes:** 29
- **Edges:** 740

Use Cases

- Peer influence analysis
- Social diffusion studies
- Classroom community detection

Overall Comparison

Dataset	Domain	Layers	Nodes	Edges
CS-Aarhus	Social	5	61	620
Drosophila	Biological	2	1363	3059
Homo	Biological	3	3471	23634
Kapferer Tailor Shop	Sociology	4	39	1018
Pierre Auger	Scientific Collaboration	16	514	7153
Sacchpomb	Biological	3	811	8363
Vickers-Chan	School Social Network	3	29	740

Important Observation

These datasets are ideal for:

- Multilayer network analysis
- Influence maximization
- Centrality algorithms
- Diffusion models (SIR/SIS/IC)
- Community detection
- GNN / Graph ML research

Especially relevant to your current work on:

- influential node detection
- multiplex networks
- diffusion dynamics
- graph decomposition methods